Q1 For the network shown below, determine

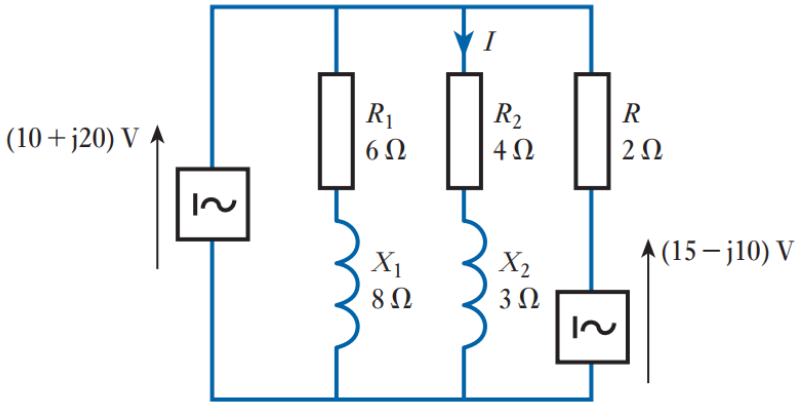


Fig. Q1

a) The current I

b) The potential difference of the inductor X_2

c) The potential difference the resistor R

Q2 Determine the Thevenin equivalent circuit for the network shown below.

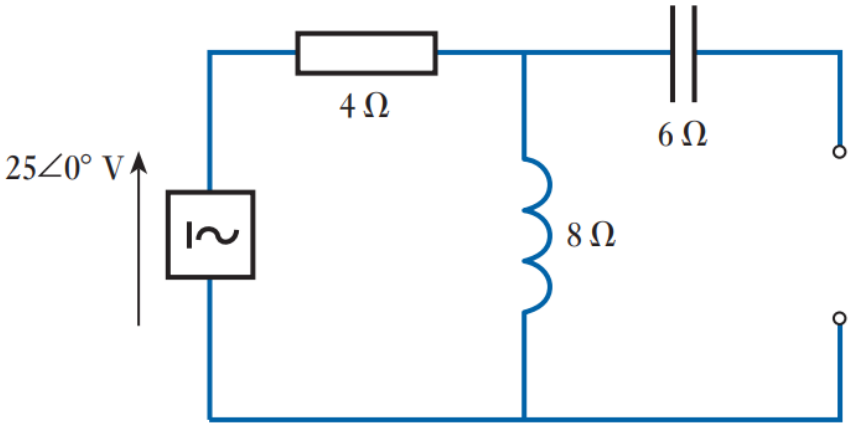


Fig. Q2

Convert the Thevenin equivalent circuit to Norton equivalent circuit.

Q3 Determine the Thevenin equivalent circuit for the network shown below.

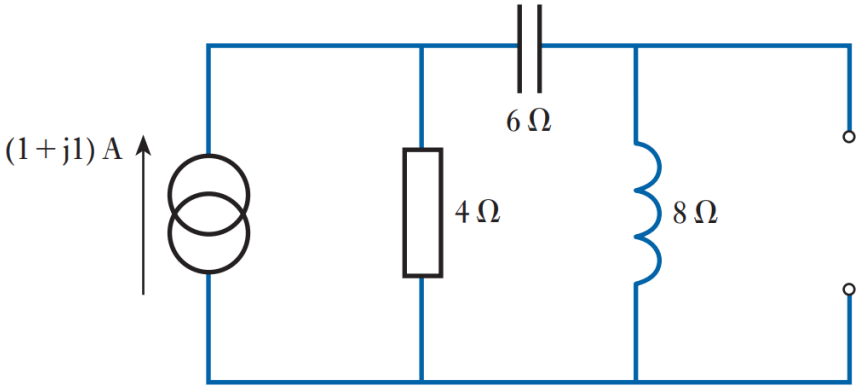


Fig. Q3

Convert the Thevenin equivalent circuit to Norton equivalent circuit.

Q4 Find the voltage gain in dB of a system when the applied signal is $3\ mV$ and the output voltage is $1.5\ V$.

Q5 An amplifier increases the power of its input from $1\ pW$ to $316.228\ nW$. Calculate the power gain.

Q6 A low-pass RC filter comprises a resistance of $159.2\ \Omega$ and a capacitor of $100\ nF$.

a) Find the cut-off frequency

b) Draw the frequency response of the filter circuit by calculating the magnitude and phase of the gain at frequencies of 1, 5, 10, 50 and $100\ kHz$.

End